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# EXPERIMENTAL TRADING PLATFORM: SCREENSHOT DOCUMENTATION

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## 1 – Consent Form

This page introduces participants to the study. It explains the purpose of the research, what participation involves, and how personal data will be used and protected. Participants are informed about their rights, compensation, confidentiality, and the ethical approval of the study. They must acknowledge that they have read and understood the information before proceeding.

Dear Participant,

You are invited to take part in an economic experiment, which is part of a project studying retail trading platforms funded by the Social Sciences and Humanities Research Council of Canada. The purpose of this study is to understand how people use data when making investment choices. The session is expected to take up to 25 minutes.

Your participation in this study is entirely voluntary. Participation presents no known risks or burdens. You will receive a compensation for participating in the study, announced on the Prolific platform, as well as a monetary compensation proportional to the profits you earn both in the process of trading and for the completion of the non-trading task. You can withdraw from the study at any time without penalty. However, if you withdraw before the end of the experiment the compensation will be forfeit, as per Prolific's rules.

Please be assured that all data generated during this study will remain confidential. You will be asked to provide some basic demographic information about yourself (gender, age, nationality, education, trading experience), but none of this information will in any way be used to personally identify you and no personal information will appear in any published study.

Your decision to participate in this study will remain completely confidential and only the researchers will have access to any of the individual data. We do not collect any data that can personally identify you. Any demographics data will be stored electronically on a web-based server until it is downloaded for the purpose of analysis. This server will only be accessible by the research team using a secure password. Data will be retained in de-identifiable format for 7 years and discarded afterwards. At that time, electronic files will be deleted and written files will be shredded. This is a standard practice for the American Psychological Association. As required by the guidelines of American Economic Association, during the publication process the data might be shared with other investigators for re-analysis. However, to ensure security and confidentiality of the data files only details of the computations sufficient to permit replication might be shared.

This study has been reviewed by the Research Ethics Board at the University of Calgary. If you are interested, you can receive a summary of the research study by emailing the primary researchers:

- Prof. Marius Zoican at [marius.zoican@ucalgary.ca](mailto:marius.zoican@ucalgary.ca)
- Prof. Mariana Khapko at [mariana.khapko@utoronto.ca](mailto:mariana.khapko@utoronto.ca)
- Dr. Philipp Chapkovski at [filipp.chapkovskii@uni-due.de](mailto:filipp.chapkovskii@uni-due.de)
- Ms. Arzu Isik at [arzu.isiktopbas@ucalgary.ca](mailto:arzu.isiktopbas@ucalgary.ca)

The research study you are participating in may be reviewed for quality assurance to make sure that the required laws and guidelines are followed. If chosen, (a) representative(s) of the Human Research Ethics Program (HREP) may access study-related data and/or consent materials as part of the review. All information accessed by the HREP will be upheld to the same level of confidentiality that has been stated by the research team.

If you wish to obtain further information regarding your rights as a participant you may contact the Conjoint Faculties Research Ethics Board (CFREB) at University of Calgary at [cfreb@ucalgary.ca](mailto:cfreb@ucalgary.ca) or by phone at 403-220-3782.

You may also contact Prof. Marius Zoican ([marius.zoican@ucalgary.ca](mailto:marius.zoican@ucalgary.ca)), Mariana Khapko ([mariana.khapko@utoronto.ca](mailto:mariana.khapko@utoronto.ca)) and Dr. Philipp Chapkovski ([filipp.chapkovskii@uni-due.de](mailto:filipp.chapkovskii@uni-due.de)) for further information or to obtain answers to any questions about this research.

I agree to participate in research being conducted by Prof. Marius Zoican and Ms. Arzu Isik of the Haskayne School of Business at University of Calgary, Prof. Mariana Khapko of the Department of Management at University of Toronto Scarborough and Dr. Philipp Chapkovski of the University Duisburg-Essen. I understand that I can withdraw my consent to participate at any time and by giving consent I am not giving up any of my legal rights.

*I have read and understood the information above* ☐

Next

Figure 1: Consent Form

## 2 – Instructions Overview

Participants receive an overview of the investment game. This section explains the core elements they will interact with—such as stock returns and order-flow imbalance—and how these variables behave throughout the trading rounds. It also outlines how trading days are structured, the role of information fees, and the decisions participants must make each day.

### Instructions

You will make a sequence of investment decisions. Read these rules carefully; your payment depends on your choices.

#### Overview

In each trading day (round), you will see a graph showing past data for a traded stock, measured over half-hour intervals.

The graph displays:

- **The stock return**, expressed in percentage terms.
- **The buy/sell imbalance (order flow imbalance)**, a measure of order flow that captures buying versus selling pressure, expressed in percentage terms.  
It is positive when buy order flow exceeds sell order flow, and negative when sell order flow exceeds buy order flow.

In about half the trading days, the buy/sell imbalance is informative and helps predict the next stock return.

In the other trading days, the two variables are independent, and the imbalance does **not** help predict returns.

#### Days and blocks

- There are 14 trading days in total: 2 training days (not paid), followed by 12 paid trading days.
- All trading days are independent, and the average stock return per 30-minute period is 15%.
- The 12 paid trading days come in three blocks of 4 days each.
- Each block is either a **Data fee** block or a **Brokerage fee** block. Block order is randomized.

#### Block conditions: Data and fees

- **Data fee block:** At the start of each day you may pay E\$4 to see buy/sell order flow imbalance data.  
If you do not pay the fee, you will only see return data.
- **Brokerage fee block:** You pay an automatic E\$4 fee each day and you can always see the buy/sell order flow imbalance data.

#### Your task each day

1. At the start of the day, you are endowed with E\$100.
2. In trading days within **Data fee** blocks, decide whether you want to purchase access to the buy/sell imbalance data.
3. Choose how much of your wealth you would like to invest in the stock. The remainder stays in cash: you cannot borrow or short-sell.
4. State whether you think the buy/sell imbalance can be used to predict the next stock return.
5. Provide a forecast for the next stock return.

#### Payment

Your final payoff comprises of four parts:

- Your portfolio value at the end of each day = cash + invested amount  $\times$  (1 + next return) – applicable fees.
- Belief bonus: E\$1 if your informative/not-informative guess is correct.
- Forecast bonus: E\$1 if your forecast is within  $\pm 1\%$  of the realized return.
- Financial quiz bonus: E\$0.25 for each correct answer in a financial literacy quiz which you will complete after the main trading days.
- One paid trading day will be selected at random for the investment component. Bonuses accumulate across paid trading days.
- Training trading days are for practice only.

After each day you will see feedback showing the realized return, whether imbalance was informative, any bonuses, fees, and the day payoff.

Next

Figure 2: Information Sheet

### **3 – Comprehension Quiz (Initial View)**

Before beginning the main task, participants complete a comprehension quiz to ensure they understand the game rules. This page presents multiple-choice questions assessing their knowledge of data-fee blocks, brokerage-fee blocks, imbalance informativeness, forecasting bonuses, and portfolio returns.

## Investment Game Comprehension Quiz

1. In a data-fee block, what is true?

- ☐ You always see the order flow data.
- ☐ You may pay E\$4 to see order flow data; otherwise, you can only see returns.
- ☐ You cannot trade in data-fee blocks.
- ☐ Order flow data is always hidden in data-fee blocks.

2. In a brokerage-fee block, what is true?

- ☐ No fees are charged and data is hidden.
- ☐ An automatic E\$4 platform fee is charged each day; order flow data is always shown.
- ☐ You must opt in each day to pay E\$4 or skip order flow data.
- ☐ Fees only apply in training days.

3. Is the buy/sell order flow imbalance always useful for predicting the next stock return?

- ☐ Yes, it is always informative.
- ☐ No, it is never informative.
- ☐ It is informative in some days and uninformative in others.
- ☐ It depends on your investment choice.

4. When do you earn the forecast bonus?

- ☐ When it is exactly equal to the realized stock return.
- ☐ When it is within  $\pm 1\%$  of the realized stock return.
- ☐ When it is within  $\pm 5\%$  of the realized stock return.
- ☐ When it is equal to the average stock return of 15%.

5. What happens to the portion of your endowment that you do not invest in the stock?

- ☐ It earns interest.
- ☐ It is lost.
- ☐ It remains in cash and stays unchanged.
- ☐ It is automatically invested for you.

6. How is the portfolio return paid?

- ☐ All 12 paid days are paid in full.
- ☐ One paid day is randomly selected; bonuses accumulate across paid days.
- ☐ Only the last day is paid.
- ☐ Only training days are paid.

Next

Figure 3: Comprehension Quiz

## 4 – Comprehension Quiz With Incorrect Answers Highlighted

After answering the quiz, participants receive immediate feedback. Incorrect answers are marked, prompting them to review the instructions and correct misunderstandings before moving on. This ensures all players fully grasp the mechanics of the experiment.

### Investment Game Comprehension Quiz

1. In a data-fee block, what is true?

- ☐ You always see the order flow data.
- ☒ You may pay E\$4 to see order flow data; otherwise, you can only see returns.
- ☐ You cannot trade in data-fee blocks.
- ☐ Order flow data is always hidden in data-fee blocks.

Please check your answer.

2. In a brokerage-fee block, what is true?

- ☒ No fees are charged and data is hidden.
- ☐ An automatic E\$4 platform fee is charged each day; order flow data is always shown.
- ☐ You must opt in each day to pay E\$4 or skip order flow data.
- ☐ Fees only apply in training days.

Please check your answer.

3. Is the buy/sell order flow imbalance always useful for predicting the next stock return?

- ☐ Yes, it is always informative.
- ☒ No, it is never informative.
- ☐ It is informative in some days and uninformative in others.
- ☐ It depends on your investment choice.

Please check your answer.

4. When do you earn the forecast bonus?

- ☐ When it is exactly equal to the realized stock return.
- ☐ When it is within  $\pm 1\%$  of the realized stock return.
- ☒ When it is within  $\pm 5\%$  of the realized stock return.
- ☐ When it is equal to the average stock return of 15%.

Please check your answer.

5. What happens to the portion of your endowment that you do not invest in the stock?

- ☐ It earns interest.
- ☒ It is lost.
- ☐ It remains in cash and stays unchanged.
- ☐ It is automatically invested for you.

6. How is the portfolio return paid?

- ☐ All 12 paid days are paid in full.
- ☒ One paid day is randomly selected; bonuses accumulate across paid days.

Figure 4: Comprehension Quiz With Feedback

## 5 – Personal Finance Self-Assessment Survey

Participants complete a short survey to assess their self-perceived financial knowledge. They rate themselves on a scale from 0 to 10, with 0 indicating no knowledge and 10 indicating high financial literacy. This measure can be used to study differences in decision-making across skill levels.

### Survey

On a scale from zero to ten, where zero is not at all knowledgeable about personal finance and ten is very knowledgeable about personal finance, what number would you be on the scale?

- ☐ 0
- ☐ 1
- ☐ 2
- ☐ 3
- ☐ 4
- ☐ 5
- ☐ 6
- ☐ 7
- ☐ 8
- ☐ 9
- ☐ 10

Next

Figure 5: Personal Finance Self-Assessment Survey



## 6 – Training Day: Brokerage Fee Acknowledgment

This screen appears on a training day and informs participants that a brokerage fee will be deducted from their daily endowment. They must acknowledge and accept the fee before accessing the trading platform. Training days do not count toward actual earnings.

5%

Instructions

### Pay Brokerage Fee

Training day: outcomes do not count toward payment.

To access the trading platform this day, a brokerage fee of E\$4 will be automatically deducted from your E\$100 endowment at the end of the day.

☐ I understand and accept this brokerage fee.

Next

Figure 6: Brokerage Fee Acknowledgment (Training)

## 7 – Training Day Trading Interface

Participants see the main trading interface for a training round. It presents historical return data and buy/sell pressure over the course of a simulated trading day. The participant chooses how much to invest, whether they believe order-flow imbalance predicts the next return, and what their forecast for the next return is. This practice round helps them understand how decisions affect outcomes.

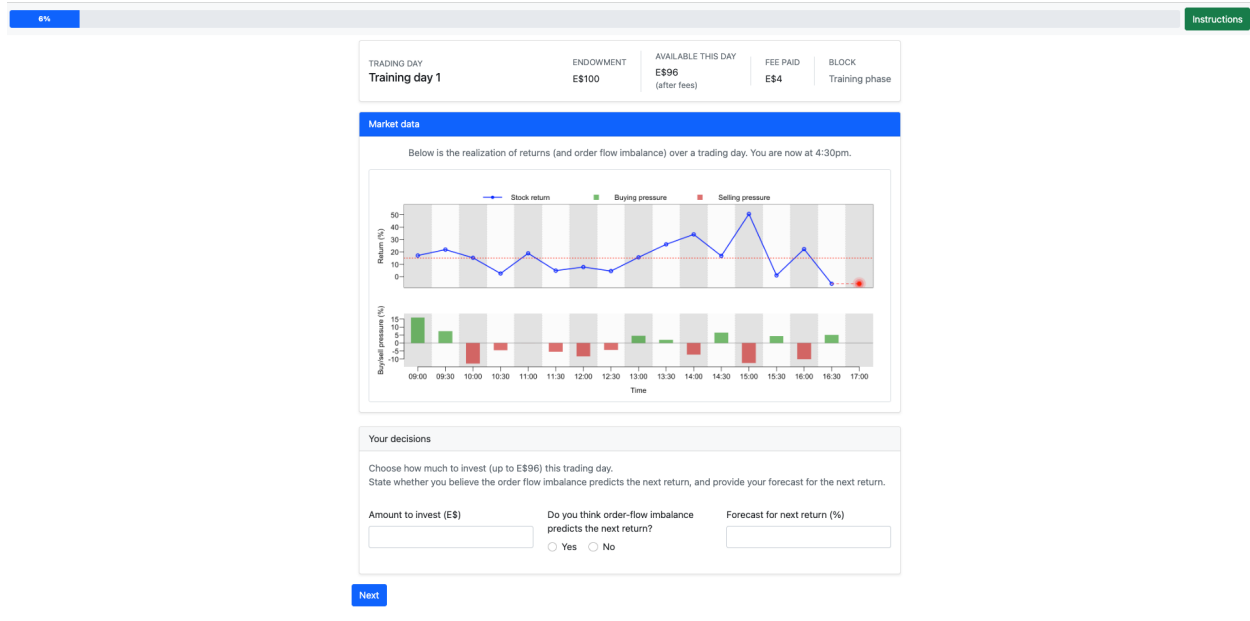


Figure 7: Training Day Trading Interface

## 8 – Training Day Results

Participants receive feedback on their training-day performance. The results page shows their investment decisions, whether the imbalance was truly informative, the realized return, and any hypothetical bonuses. Fees and payoffs are displayed for transparency, though training outcomes do not affect actual earnings.

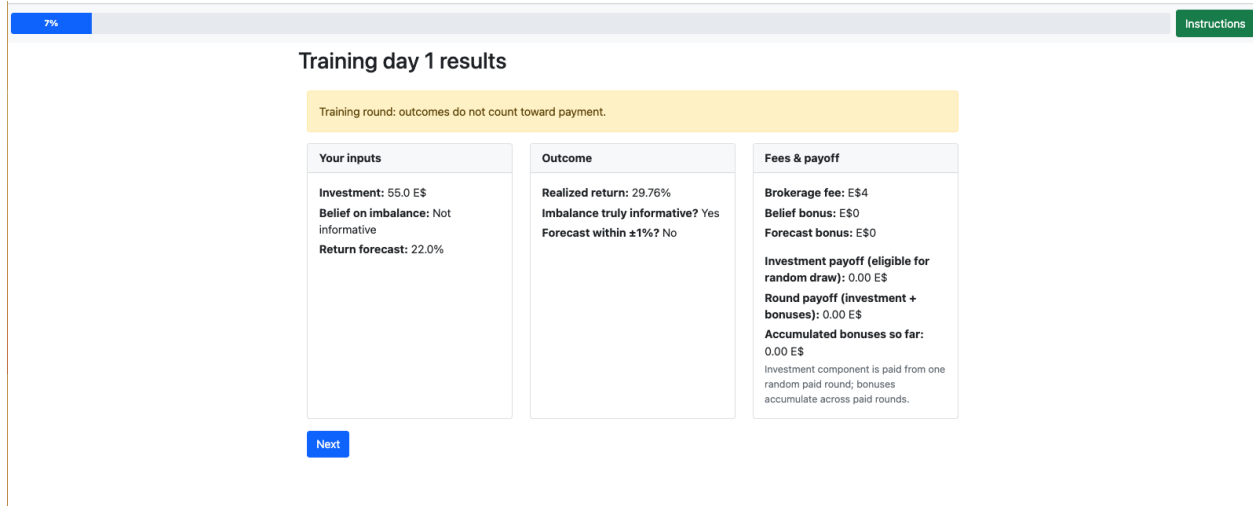


Figure 8: Training Day Results

## 9 – Beginning of Block 1

Participants are informed that the paid portion of the experiment is about to begin. The block structure (twelve paid trading days) and the two fee regimes—brokerage-fee blocks and data-fee blocks—are explained. Participants are reminded that order-flow imbalance is informative only on some days.

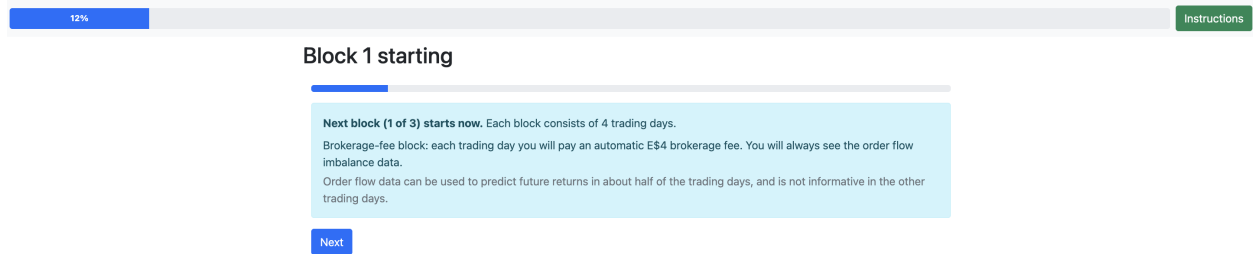


Figure 9: Beginning of Paid Part of the Experiment

## 10 – Treatment: Data-Fee Block (Data Purchase Decision)

On data-fee days, participants must decide whether to pay a fixed fee to access order-flow imbalance data. This screen informs them that the current day is in a data-fee block and asks whether they want to pay E\$4 for data. Choosing “Yes” reveals the imbalance series for the day; choosing “No” keeps the platform identical but hides the order-flow imbalance.

21%

Instructions

### Purchase Order Flow Imbalance Data

This day is in a data-fee block. You may pay E\$4 to see the data for this day.

**Do you want to pay E\$4 for data this day?**

☒ Yes

☐ No

Next

Figure 10: Data-Fee Block: Choice to Pay for Order-Flow Imbalance Data

## 11 – Treatment: Data-Fee Block (Choice Not to Pay)

This variant of the data-choice screen illustrates the case where a participant selects “No.” The interface confirms that they have chosen not to pay E\$4 for data on this day. They proceed to the trading platform without access to order-flow imbalance.

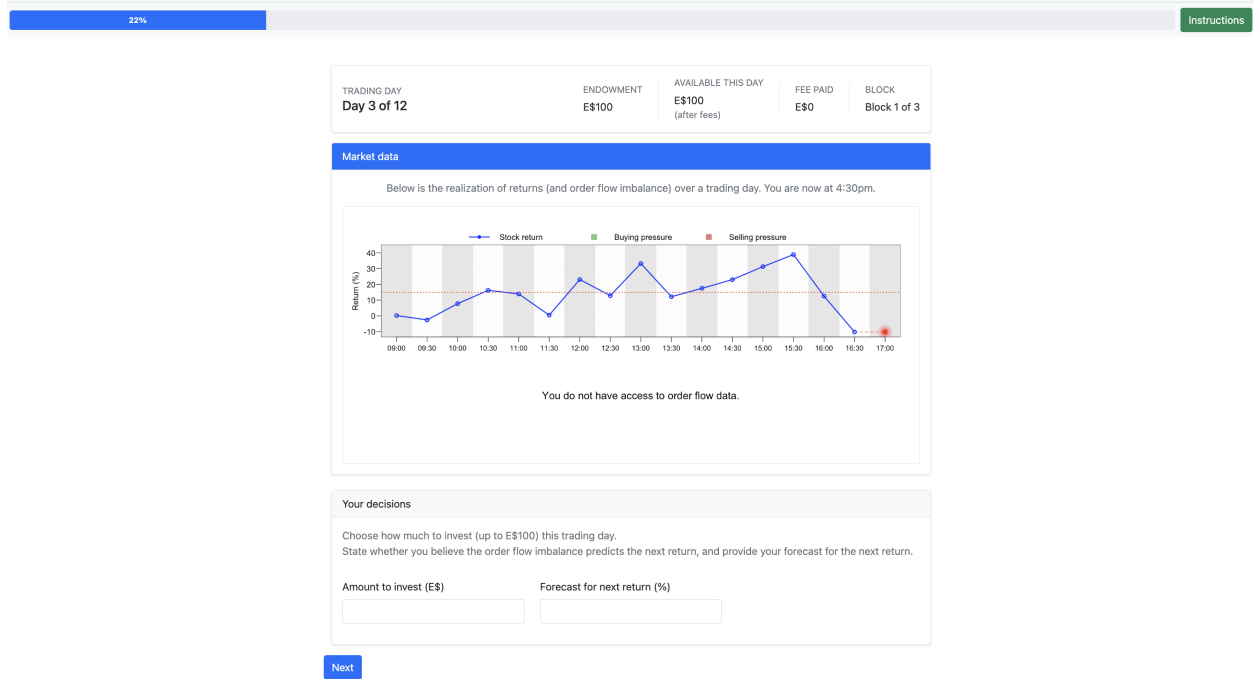


Figure 11: Data-Fee Block: Choosing Not to Purchase Data

## 12 – Paid Trading Platform With Data

When participants either pay the data fee (in a data-fee block) or are in a brokerage-fee block, the trading interface displays both stock returns and order-flow imbalance. The upper panel tracks the intraday price path, while the colored bars indicate buying and selling pressure. Participants decide how much to invest, whether the imbalance predicts the next return, and their forecast for the next period.

The screenshot shows a trading interface with a progress bar at the top indicating 25% completion. A green 'Instructions' button is located in the top right corner. The main heading is 'Purchase Order Flow Imbalance Data'. Below this, a text box contains the message: 'This day is in a data-fee block. You may pay E\$4 to see the data for this day.' Below the text box, a question is posed: 'Do you want to pay E\$4 for data this day?'. There are two radio button options: 'Yes' and 'No'. The 'No' option is selected, indicated by a blue dot. At the bottom left of the form, there is a blue 'Next' button.

25%

Instructions

### Purchase Order Flow Imbalance Data

This day is in a data-fee block. You may pay E\$4 to see the data for this day.

Do you want to pay E\$4 for data this day?

☐ Yes

☒ No

Next

Figure 12: Paid Trading Interface When Order-Flow Imbalance Data Are Visible

### 13 – Paid Trading Platform Without Data

If a participant chooses not to pay the data fee on a data-fee day, the trading interface is identical except that order-flow imbalance is hidden. The return path is still displayed, but the imbalance series and its corresponding legend are removed. Participants must then make investment and forecasting decisions based only on past returns, allowing the experiment to measure the value they place on access to order-flow data.

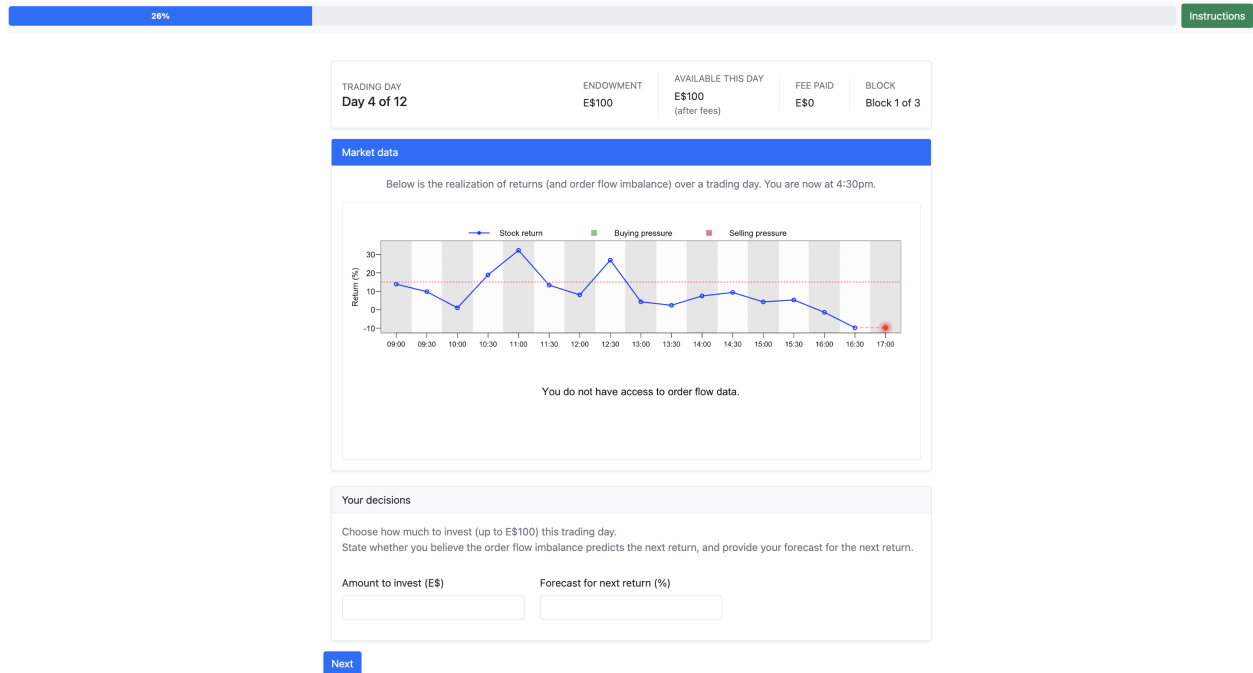
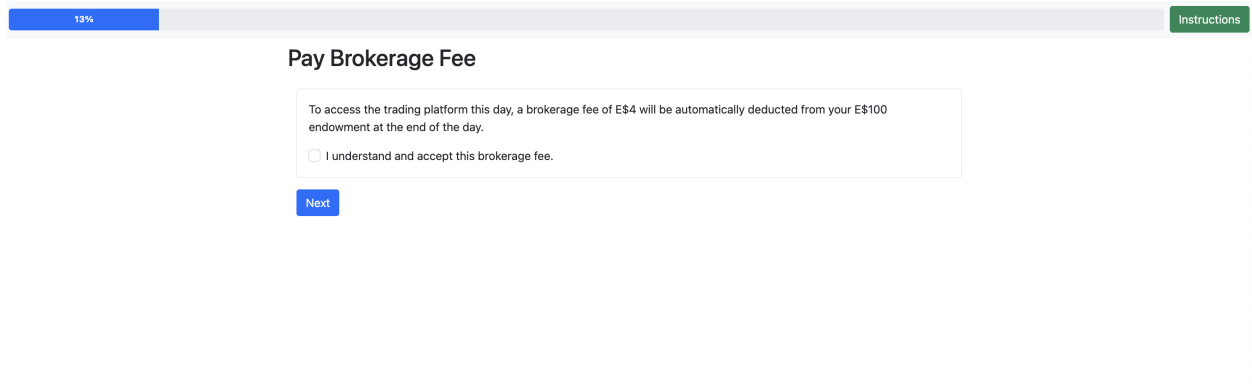


Figure 13: Paid Trading Interface When Order-Flow Imbalance Data Are Not Purchased



## 14 – Example Paid Day: Brokerage/Data Fee Confirmation

At the start of each paid trading day, participants are reminded of the applicable fee regime. The screen shows either a brokerage fee that will be automatically deducted, or the opportunity to pay a data fee (as shown above). They must confirm this information before continuing.



The screenshot shows a web interface for a trading platform. At the top, there is a progress bar with a blue segment labeled '13%' and a green segment labeled 'Instructions'. Below this, the main heading is 'Pay Brokerage Fee'. A text box contains the following information: 'To access the trading platform this day, a brokerage fee of E\$4 will be automatically deducted from your E\$100 endowment at the end of the day.' Below this text is a radio button followed by the text 'I understand and accept this brokerage fee.' At the bottom left of the form area is a blue button labeled 'Next'.

Figure 14: Example Fee Confirmation Screen for a Paid Day

## 15 – Example Paid Day: Trading Interface

This screenshot illustrates one typical paid trading day. The participant observes the series of intraday stock returns (and, when applicable, order-flow imbalance), chooses an investment amount up to their available endowment, states whether they believe imbalance predicts the next return, and enters a numerical forecast for the next period's return.

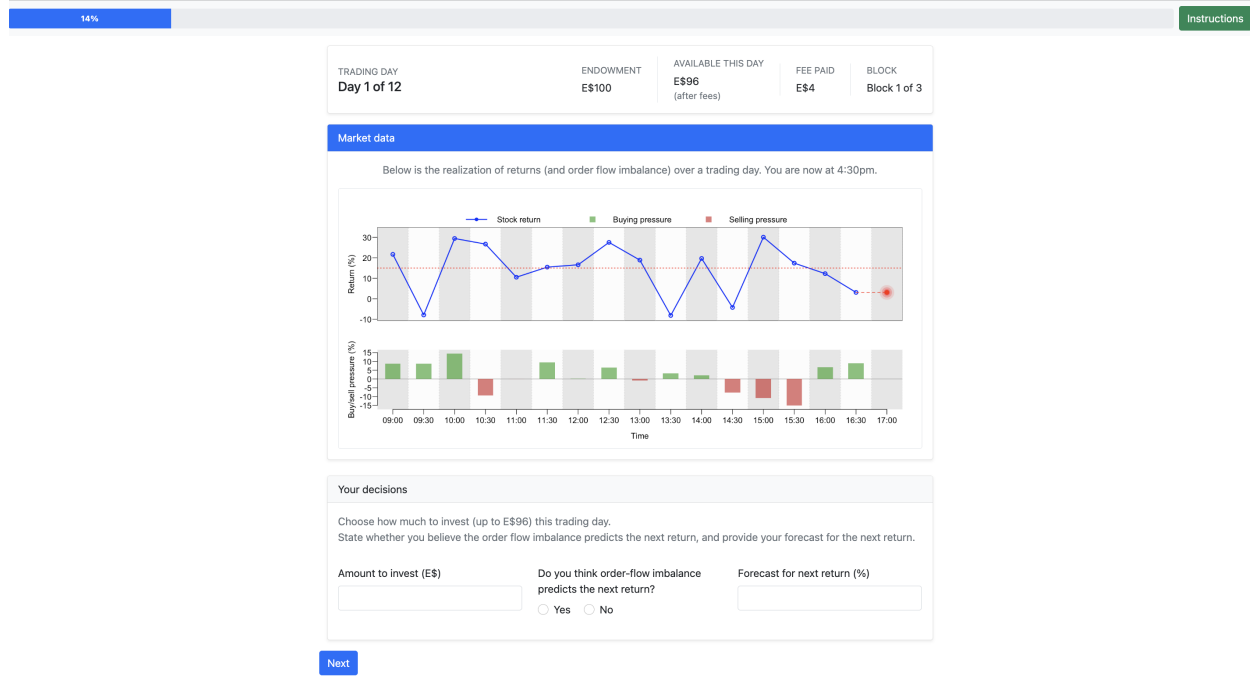


Figure 15: Example Paid Trading Interface

## 16 – Example Paid Day: Results

After submitting their decisions, participants see the outcome for that day. The results page reports the realized return, whether order-flow imbalance was truly informative, how accurate their forecast was, and their payoff net of fees. One of the paid rounds is later selected at random to determine the investment component of their final earnings.

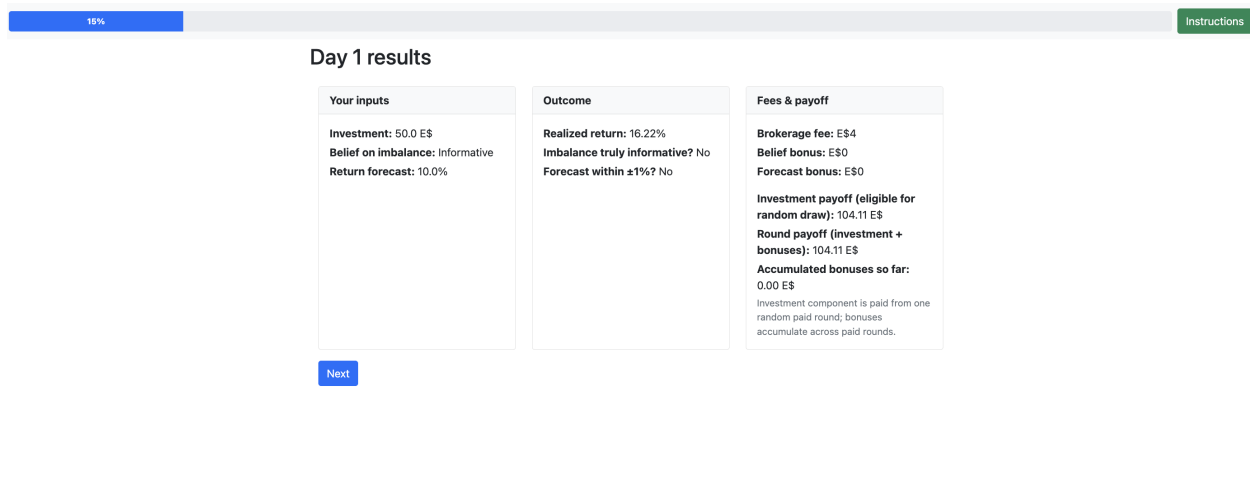


Figure 16: Example Paid Day Results

## **17 – Remaining Paid Rounds (Summary)**

The experiment continues for a total of twelve trading days. The structure of the remaining rounds is identical to the example shown above: each day begins with a fee confirmation or data-purchase screen, followed by the trading interface and a results page. To avoid unnecessary repetition, screenshots for Days 2–12 are not included here. Instead, this documentation focuses on:

- consent and instructions,
- the comprehension quiz and training rounds,
- the data-fee treatment (choice to purchase order-flow data),
- the trading platform with and without order-flow data, and
- one representative paid trading day with its results.

## 18 – Holt–Laury Risk Preference Check

This screen presents the Holt–Laury risk preference task, a standard economic measure of individual risk aversion. Participants choose between paired lotteries, labeled Option A and Option B, across ten rows with progressively increasing probabilities of the higher payoff. They are instructed to select exactly one option per row and to switch from the safer Option A to the riskier Option B no more than once. This switching point provides a measure of their risk tolerance.

99%

**Risk Preference Check**

**Holt-Laury risk choices** Select exactly one option per row

Each row shows two lotteries. Option A pays E\$2.00 with the stated probability and E\$1.60 otherwise. Option B pays E\$3.85 with the stated probability and E\$0.10 otherwise. Please indicate the first row at which you would switch from Option A to Option B. Choices must switch at most once (A→B).

| Row | Option A                                              | Option B                                                 |
|-----|-------------------------------------------------------|----------------------------------------------------------|
| 1   | 1/10 of \$2 / 9/10 of \$1.6<br><input type="radio"/>  | 1/10 of \$3.85 / 9/10 of \$0.1<br><input type="radio"/>  |
| 2   | 2/10 of \$2 / 8/10 of \$1.6<br><input type="radio"/>  | 2/10 of \$3.85 / 8/10 of \$0.1<br><input type="radio"/>  |
| 3   | 3/10 of \$2 / 7/10 of \$1.6<br><input type="radio"/>  | 3/10 of \$3.85 / 7/10 of \$0.1<br><input type="radio"/>  |
| 4   | 4/10 of \$2 / 6/10 of \$1.6<br><input type="radio"/>  | 4/10 of \$3.85 / 6/10 of \$0.1<br><input type="radio"/>  |
| 5   | 5/10 of \$2 / 5/10 of \$1.6<br><input type="radio"/>  | 5/10 of \$3.85 / 5/10 of \$0.1<br><input type="radio"/>  |
| 6   | 6/10 of \$2 / 4/10 of \$1.6<br><input type="radio"/>  | 6/10 of \$3.85 / 4/10 of \$0.1<br><input type="radio"/>  |
| 7   | 7/10 of \$2 / 3/10 of \$1.6<br><input type="radio"/>  | 7/10 of \$3.85 / 3/10 of \$0.1<br><input type="radio"/>  |
| 8   | 8/10 of \$2 / 2/10 of \$1.6<br><input type="radio"/>  | 8/10 of \$3.85 / 2/10 of \$0.1<br><input type="radio"/>  |
| 9   | 9/10 of \$2 / 1/10 of \$1.6<br><input type="radio"/>  | 9/10 of \$3.85 / 1/10 of \$0.1<br><input type="radio"/>  |
| 10  | 10/10 of \$2 / 0/10 of \$1.6<br><input type="radio"/> | 10/10 of \$3.85 / 0/10 of \$0.1<br><input type="radio"/> |

Please choose A in early rows and switch to B at most once; no A choices after switching to B. Also select an option for every row.

Detected switch point: none yet

Next

Figure 17: Holt–Laury Risk Preference Check

## 19 – Financial Quiz: Compound Interest Question

Participants begin a financial literacy assessment. The first question tests understanding of compound interest: given a savings account earning 3% annually over four years, they must determine whether the ending balance would exceed, equal, or fall below \$112. This evaluates basic numeracy and comprehension of interest accumulation.

98%

### Financial Quiz

Suppose you had \$100 in a savings account and the interest rate was 3% per year. After 4 years, how much do you think you would have in the account if you left the money to grow? \*

☐ More than \$112

☐ Exactly \$112

☐ Less than \$112

☐ Don't know

☐ Prefer not to say

Next

Figure 18: Financial Quiz: Compound Interest Question

## 20 – Financial Quiz: Real Interest Rate / Inflation Question

This item assesses understanding of real purchasing power. Participants are asked whether money in an account earning 1% interest, while inflation is 2%, would allow them to buy more, the same, or less after one year. The question evaluates awareness of how inflation erodes real returns.

96%

### Financial Quiz

Imagine that the interest rate on your savings account was 1% per year and inflation was 2% per year. After 1 year, with the money in this account, would you be able to buy \*

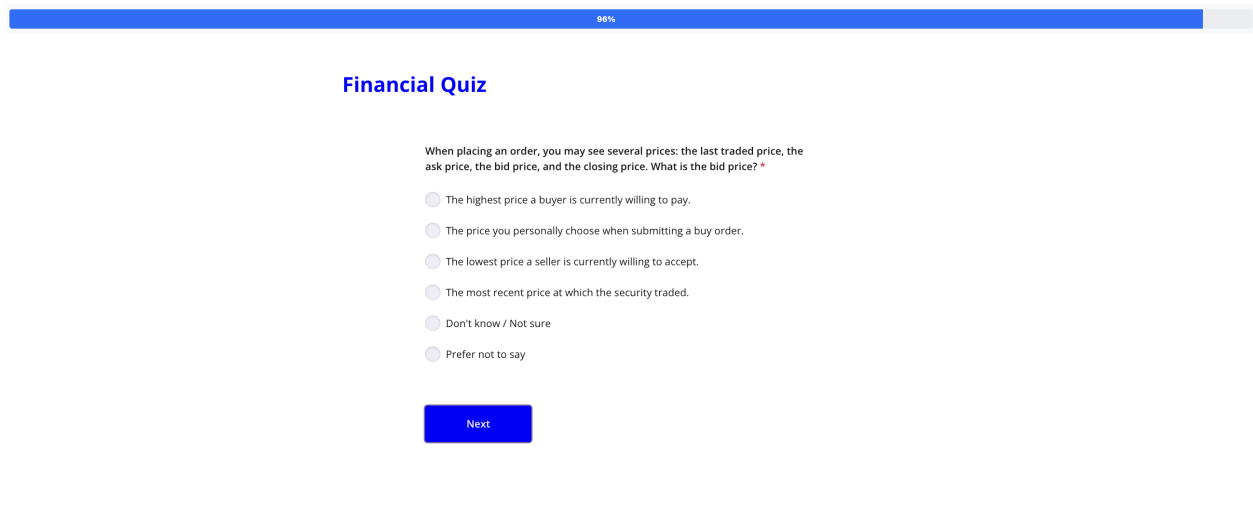
- ☐ More than today
- ☐ Exactly the same as today
- ☐ Less than today
- ☐ Don't know / Not sure
- ☐ Prefer not to say

Next

Figure 19: Financial Quiz: Real Interest Rate / Inflation Question

## 21 – Financial Quiz: Bid Price Definition

Participants answer a conceptual question about market microstructure. They must identify the correct definition of the bid price from several options, demonstrating understanding of trading mechanics and terminology relevant to securities markets.



The screenshot displays a quiz interface within a blue-bordered container. At the top, a blue progress bar shows 96% completion. Below the bar, the title "Financial Quiz" is centered in blue. The question text reads: "When placing an order, you may see several prices: the last traded price, the ask price, the bid price, and the closing price. What is the bid price? \*". There are six radio button options listed below the question. A blue "Next" button is positioned at the bottom of the options.

96%

### Financial Quiz

When placing an order, you may see several prices: the last traded price, the ask price, the bid price, and the closing price. What is the bid price? \*

- ☐ The highest price a buyer is currently willing to pay.
- ☐ The price you personally choose when submitting a buy order.
- ☐ The lowest price a seller is currently willing to accept.
- ☐ The most recent price at which the security traded.
- ☐ Don't know / Not sure
- ☐ Prefer not to say

Next

Figure 20: Financial Quiz: Bid Price Definition



## 22 – Financial Quiz: Mortgage Structure

This question evaluates comprehension of borrowing costs. Participants must determine whether a 15-year mortgage, despite requiring higher monthly payments than a 30-year mortgage, results in lower total interest paid over the life of the loan. This tests financial reasoning related to credit and long-term debt.

98%

### Financial Quiz

A 15-year mortgage typically requires higher monthly payments than a 30-year mortgage, but the total interest paid over the life of the loan will be less. \*

☐ True

☐ False

☐ Don't know / Not sure

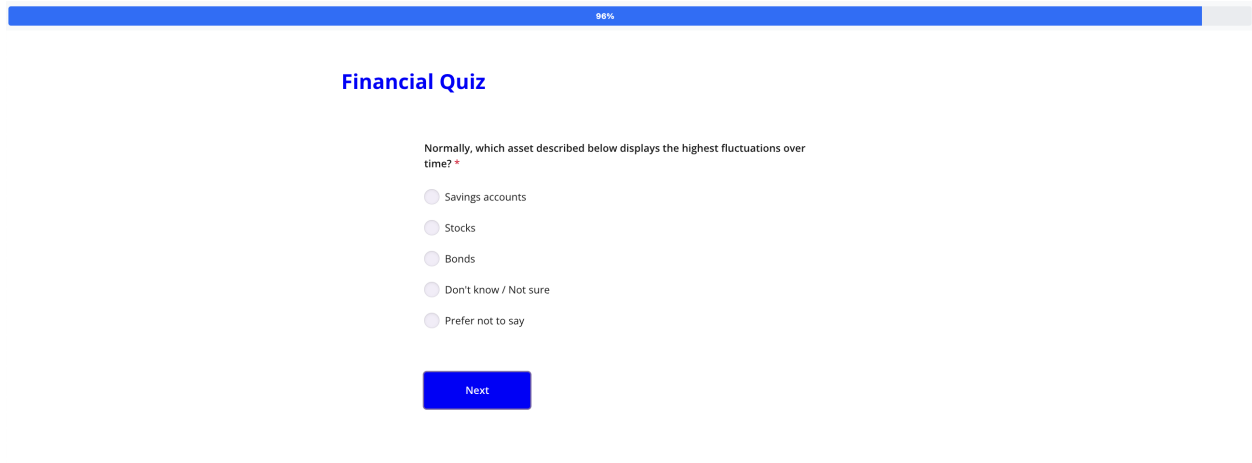
☐ Prefer not to say

Next

Figure 21: Financial Quiz: Mortgage Structure

## 23 – Financial Quiz: Asset Volatility

Participants identify which asset class typically exhibits the highest price fluctuations over time. This question distinguishes between low-volatility assets (such as savings accounts or bonds) and higher-risk investments (such as stocks), assessing understanding of risk–return tradeoffs.



The screenshot shows a quiz interface with a blue progress bar at the top indicating 96% completion. The title "Financial Quiz" is displayed in blue. The question text is "Normally, which asset described below displays the highest fluctuations over time?" followed by a red asterisk. There are five radio button options: "Savings accounts", "Stocks", "Bonds", "Don't know / Not sure", and "Prefer not to say". A blue "Next" button is located at the bottom of the question area.

96%

### Financial Quiz

Normally, which asset described below displays the highest fluctuations over time? \*

- ☐ Savings accounts
- ☐ Stocks
- ☐ Bonds
- ☐ Don't know / Not sure
- ☐ Prefer not to say

Next

Figure 22: Financial Quiz: Asset Volatility

## 24 – Financial Quiz: Diversification

This item measures understanding of diversification. Participants evaluate whether spreading investments across multiple assets increases, decreases, or leaves unchanged the risk of losing large amounts of money. This is a key concept in portfolio theory and financial risk management.

96%

### Financial Quiz

When an investor spreads his money among different assets, does the risk of losing a lot of money <sup>a</sup>

- ☐ Increase
- ☐ Stay the same
- ☐ Decrease
- ☐ Don't know / Not sure
- ☐ Prefer not to say

Next

Figure 23: Financial Quiz: Diversification

## 25 – Financial Quiz: Long-Term Returns

Participants are asked which asset normally yields the highest return over long horizons—typically stocks, compared to low-risk alternatives such as savings accounts or bonds. This assesses knowledge of historical risk–return patterns in financial markets.

96%

### Financial Quiz

Considering a long time period (for example, 10 or 20 years), which asset described below normally gives the highest return? \*

☐ Savings accounts

☐ Stocks

☐ Bonds

☐ Don't know / Not sure

☐ Prefer not to say

Next

Figure 24: Financial Quiz: Long-Term Returns

## 26 – Financial Quiz: Mutual Fund Risk

This question tests whether participants understand that investing in a stock mutual fund involves uncertainty and that the value may fall below the initial amount invested. It probes awareness of market volatility and downside risk.

96%

### Financial Quiz

Do you think that the following statement is true or false? "If you were to invest \$1,000 in a stock mutual fund, it would be possible to have less than \$1,000 when you withdraw your money." \*

☐ True

☐ False

☐ Don't know / Not sure

☐ Prefer not to say

Next

Figure 25: Financial Quiz: Mutual Fund Risk

## 27 – Financial Quiz: Bid–Ask Spread

The final quiz item in this block focuses again on market microstructure. Participants are asked to identify the correct meaning of the bid–ask spread, which is the difference between the highest price a buyer is willing to pay (bid) and the lowest price a seller is willing to accept (ask). This measures understanding of trading frictions and liquidity.

99%

### Financial Quiz

When viewing a security's quote, you may see both a bid price and an ask price.  
What is the bid–ask spread? \*

- ☐ The average of the bid price and the ask price.
- ☐ The difference between the lowest ask price and the highest bid price.
- ☐ The fee the broker charges each time you trade.
- ☐ The change in the stock price from market open to market close.
- ☐ Don't know / Not sure
- ☐ Prefer not to say

Next

Figure 26: Financial Quiz: Bid–Ask Spread

## 28 – Financial Quiz: Mutual Funds

This question assesses participants' understanding of how mutual funds operate. They must identify which statement is correct, testing knowledge of fund structure, asset diversification, and the misconception that mutual funds guarantee returns. The item evaluates foundational financial literacy regarding investment vehicles.

98%

### Financial Quiz

Which of the following statements is correct? \*

- ☐ Once one invests in a mutual fund, one cannot withdraw the money in the first year
- ☐ Mutual funds can invest in several assets, for example invest in both stocks and bonds
- ☐ Mutual funds pay a guaranteed rate of return which depends on their past performance
- ☐ None of the above
- ☐ Don't know / Not sure
- ☐ Prefer not to say

Next

Figure 27: Financial Quiz: Mutual Funds

## 29 – Financial Quiz: Bonds

This question examines the participant's understanding of corporate bonds. They must choose the correct interpretation of what it means to buy a bond from firm B. The correct concept—that a bond represents lending money to the firm—distinguishes debt financing from equity ownership. This item tests clarity on financial instruments and investor rights.

96%

### Financial Quiz

Which of the following statements is correct? If somebody buys a bond of firm B \*

- ☐ She owns a part of firm B
- ☐ She is liable for the company's debts
- ☐ She has lent money to firm B
- ☐ None of the above
- ☐ Prefer not to say

Next

Figure 28: Financial Quiz: Bonds



### 30 – Financial Quiz: Credit Card Debt

This question evaluates financial reasoning related to consumer debt. Given a credit card balance, minimum payment, and annual interest rate, participants must estimate how long it would take to repay the debt without additional charges. It tests awareness of how compounding interest prolongs repayment, highlighting the consequences of only making minimum payments.

The screenshot shows a quiz interface with a blue progress bar at the top indicating 96% completion. The title 'Financial Quiz' is displayed in blue. The question text is: 'Suppose you owe \$3,000 on your credit card. You pay a minimum payment of \$30 each month. At an annual percentage rate of 12% (or 1% per month), how many years would it take to eliminate your credit card debt if you made no additional new charges? \*'. Below the question are five radio button options: 'Less than 5 years', 'Between 5 and 10 years', 'Between 10 and 15 years', 'Never', and 'None of the above'. A 'Prefer not to say' option is also present. A blue 'Complete' button is located at the bottom of the quiz area.

96%

#### Financial Quiz

Suppose you owe \$3,000 on your credit card. You pay a minimum payment of \$30 each month. At an annual percentage rate of 12% (or 1% per month), how many years would it take to eliminate your credit card debt if you made no additional new charges? \*

☐ Less than 5 years

☐ Between 5 and 10 years

☐ Between 10 and 15 years

☐ Never

☐ None of the above

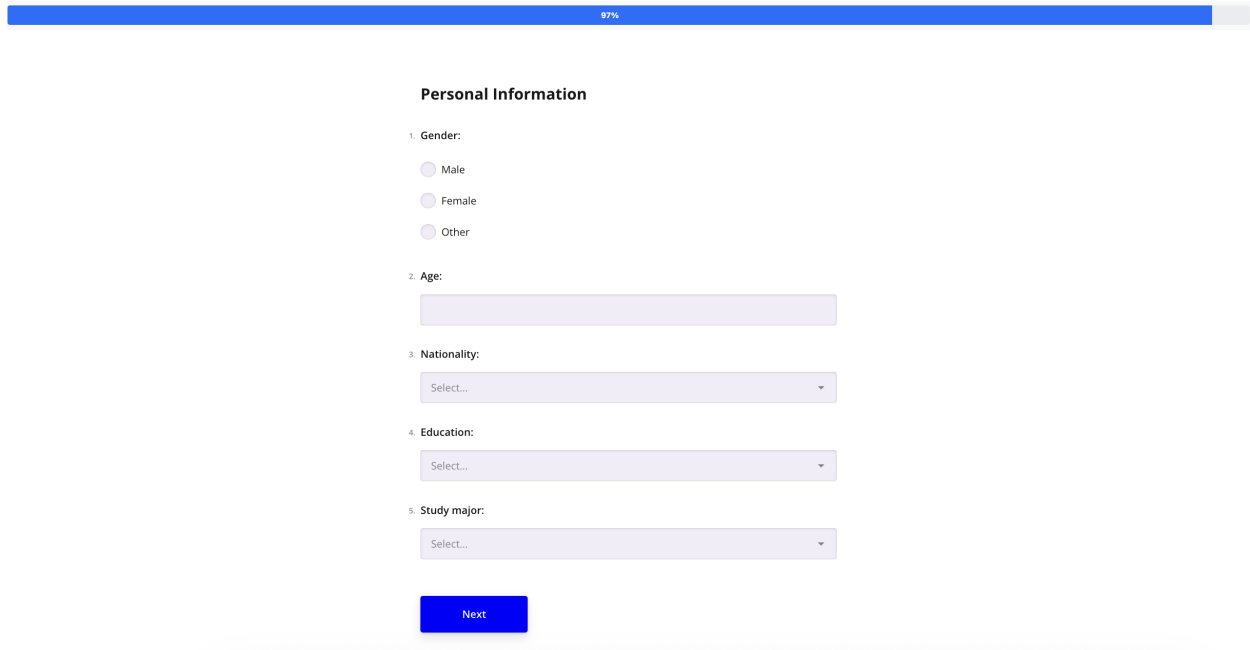
☐ Prefer not to say

Complete

Figure 29: Financial Quiz: Credit Card Debt

## 31 – Personal Information Survey

Participants provide demographic information, including gender, age, nationality, education level, and field of study. Collecting these variables enables researchers to analyze heterogeneity in decision-making across demographic groups and control for background characteristics in statistical models.



The screenshot displays a web-based survey titled "Personal Information". At the top, a blue progress bar indicates 97% completion. The survey consists of five numbered questions:

- 1. Gender:** Three radio button options: Male, Female, and Other.
- 2. Age:** A single-line text input field.
- 3. Nationality:** A dropdown menu with "Select..." as the placeholder text.
- 4. Education:** A dropdown menu with "Select..." as the placeholder text.
- 5. Study major:** A dropdown menu with "Select..." as the placeholder text.

At the bottom of the form is a blue "Next" button.

Figure 30: Personal Information Survey

## 32 – Financial Experience Survey (Part 1)

This section gathers information about participants' prior exposure to financial markets. Questions cover coursework in finance, experience participating in experiments, real-world trading experience, use of trading apps, and frequency of online trading. These measures help researchers understand how prior knowledge and market familiarity may influence behavior in the experiment.

97%

### Financial Experience

6. Did you take any course focused on financial markets?

No Yes

7. Have you taken part in an experiment before?

No Yes

8. Do you have any trading experience?

No Yes

9. Do you use mobile trading apps?

No Yes

10. How often do you trade online?

☐ Multiple times a day

☐ Daily

☐ Weekly

☐ Monthly

☐ Less than once a month

Figure 31: Financial Experience Survey (Part 1)

### 33 – Financial Experience Survey (Part 2)

Participants continue reporting aspects of their financial activity. They indicate how often they monitor their portfolio, which asset class they trade most frequently, and whether they use leverage. These variables proxy for financial engagement and sophistication, allowing researchers to explore relationships between market experience and experimental choices.

97%

11. How often do you check the value of your portfolio?

- ☐ Multiple times a day
- ☐ Daily
- ☐ Weekly
- ☐ Monthly
- ☐ Less than once a month

12. Which asset class do you trade the most?

- ☐ Stocks
- ☐ Bonds
- ☐ Derivatives (Options, Futures)
- ☐ Cryptocurrencies

13. Do you use leverage (e.g., trading on margin)?

- ☐ Yes
- ☐ No
- ☐ Do not know

Previous Next

Figure 32: Financial Experience Survey (Part 2)

## 34 – Feedback Questions

Participants are invited to reflect on the study. They describe what they believe the purpose of the experiment is and report any difficulties encountered. These open-ended responses provide insight into participant understanding, potential misconceptions, and usability issues in the experimental interface.

The screenshot displays a feedback form titled "Feedback Questions". At the top, a blue progress bar indicates 97% completion. Below the title, there are two questions, each followed by a large text input area:

- 14. What do you think is the purpose of this study?
- 15. Did you encounter any difficulty throughout the experiment?

At the bottom of the form, there are two buttons: "Previous" (light blue) and "Next" (dark blue).

Figure 33: Feedback Questions

## 35 – Payoff Summary

This final payoff screen summarizes the participant's earnings. It reports the randomly selected trading day used for payment, the payoff from that day's investment, accumulated belief and forecast bonuses, lottery bonuses, and quiz earnings. The screen then converts total points into a real monetary bonus using a fixed exchange rate. This transparent breakdown ensures participants understand how their final compensation was determined.

### Payoff

| Payment Details                                   |
|---------------------------------------------------|
| Random trading day chosen to pay: Day 5 of 12     |
| Payoff for trade: 106.42 E\$                      |
| Accumulated belief/forecast bonuses: 7.00 E\$     |
| Payoff from lottery draw: 2.00 E\$                |
| Number of correct answers in quiz: 2/12           |
| Fee per correct answer: 0.25                      |
| Payoff for quiz: 0.50 E\$                         |
| Total points (trade + bonuses + quiz): 113.92 E\$ |
| Total points including lottery: 115.92 E\$        |
| Exchange rate: 1 point = 0.075                    |
| Cash bonus (points × rate): 8.69 E\$              |
| Participation fee: \$0.00                         |
| Your total bonus in real money: 8.69 E\$          |

Next

Figure 34: Payoff Summary

## 36 – Additional Comments and Exit

Participants are given a final opportunity to share feedback on the platform's usability and are redirected to the Prolific completion page. This supports iterative refinement of the experiment interface and ensures proper tracking, compensation, and verification of study completion.

### Comments

We value your feedback on the usability of this platform. Please share any technical issues, glitches, bugs, or suggestions for improvement to help us enhance your experience.

Next

Figure 35: Additional Comments

## 37 – Prolific Login Page

The final screen shows the Prolific login interface, where participants authenticate their identity and access the study through Prolific's participant management system. This ensures proper tracking, compensation, and verification of study completion.



Log in

Email address\*

Password\*  

[Forgot password?](#)

OR



Don't have an account? [Sign up](#)

Figure 36: Prolific Login Page